

Fields as Persistence Carriers

The Minimal Carrier Role of Physical Field Structure under Bounded Perturbation Transport

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Abstract

This paper derives the minimal carrier role of physical field structure within the realized-physics arc. Previous results established non-vacuous realized persistence, bounded standing-localization, continuation-neighborhood domains, quotient-stable comparison carriers, and bounded perturbation transport. The present paper asks what carries localized persistence through such bounded transport. Localized persistence cannot be carried by isolated points, path labels, global undifferentiated substrates, hidden reservoirs, trace-only descriptions, or bookkeeping data attached to domains after the fact. A carrier must mediate standing-bearing transport, not merely record it. Standing-bearing persistence under bounded perturbation transport therefore requires a distributed carrier assignment over local continuation domains preserving boundary trace, standing status, identity-memory, restoration status, perturbation burden, quotient comparison, and admitted transport interface.

The main result is a field-carrier role theorem. In a fixed admissible realized persistence regime with local continuation domains, a quotient comparison carrier, and bounded perturbation transport, localized persistence induces a partial field-role assignment

$$\Phi_D : \text{Dom}(\Phi_D) \subseteq \text{LCD}_D \rightarrow \text{CarProf}_D$$

whose local carrier profiles cohere under admitted restriction, boundary overlap, and perturbation transport witnesses. This assignment is field-structural in the minimal physical sense because admissible modification of its profiles changes the regime's standing-bearing transport organization; it is not a passive information layer. It is the role later occupied by scalar, vector, tensor, gauge, or quantum fields once additional formal structure is supplied. No spacetime manifold, field equation, Lagrangian, Hamiltonian, gauge bundle, operator algebra, stress-energy tensor, particle ontology, or quantum field formalism is assumed.

1 Introduction

The preceding papers establish a cumulative reconstruction ladder from admissible realized persistence toward recognizable physical roles. Minimal non-vacuity fixes the floor: realized persistence cannot be reconstruction-null inside its own admissible continuation regime [1]. Standing localization then shows that standing-bearing persistence cannot remain global, diffuse, hidden, or unbounded once perturbation and restorative continuation constraints are admitted [2]. Continuation-locus analysis derives boundary-compatible continuation-neighborhood domains [3]. Stable comparison among such domains then induces a quotient comparison carrier [4]. Bounded perturbation transport shows that standing-relevant perturbation burden cannot move globally, pathlessly, or without admitted witnesses [5].

In compact form, the chain is

$$\text{RealPers}_D \Rightarrow \text{NonVacRec}_D \Rightarrow [B_D]_{\equiv_B} \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \Rightarrow C_D \Rightarrow \text{BoundTrans}_D.$$

The present paper begins at the final term. Local continuation domains exist. Quotient-stable comparison exists. Standing-relevant perturbation transport is bounded and witness-mediated. The next physical question is unavoidable:

What carries localized persistence through bounded perturbation transport?

The answer cannot be an isolated point, a route label, a global undifferentiated medium, a hidden reservoir, a trace-only record, or bookkeeping metadata assigned after transport has already been described. Bounded perturbation transport preserves standing status, identity-memory, restoration status, boundary trace, perturbation burden, quotient comparison, and transport interface. These roles must be carried locally, must mediate the admitted transport itself, and must cohere across admitted continuation domains. The resulting structure is the minimal physical field role.

1.1 The physical role claim

The claim is not that a scalar field, vector field, tensor field, gauge potential, quantum operator field, or smooth section over a manifold is assumed. The claim is prior to those species:

Field structure first enters realized physics as the distributed carrier role required for localized persistence to retain standing-bearing consequence under bounded perturbation transport.

Standard field theories provide later formal specializations of this role. Classical field theory represents field values over spacetime regions; gauge theory adds equivalence under local transformations; quantum field theory adds operator or algebraic structure; field dynamics adds equations, actions, and conserved quantities [7, 8, 9, 10, 11, 12, 13]. None of that is imported here. The paper derives why a realized regime already requires a distributed carrier assignment whose profiles mediate localized persistence and perturbation transport before those mathematical enrichments can be supplied.

1.2 Carrier mediation, not bookkeeping

The decisive distinction is between a bookkeeping assignment and a field-role assignment. A bookkeeping assignment records where standing, burden, or transport has been described. A field-role assignment mediates the admissible preservation of standing-bearing persistence itself. In the present paper, a carrier counts as field-structural only when admissible modification of that carrier changes the standing-bearing transport organization of the regime.

Bookkeeping assignment	Field-role assignment
labels persistence	carries persistence
records a route or state	mediates transport
can change without changing admissible transport	changes transport organization when altered
external description	domain-internal role support
syntax or presentation skin	standing-bearing carrier structure

This distinction prevents the central theorem from being read as a claim about distributed data. The theorem concerns distributed carrier mediation. Field structure enters because bounded perturbation transport requires a role-bearing support that preserves standing, identity, restoration, burden, boundary, quotient, and transport interface across local domains.

1.3 Main result and scope

The central implication is

$$(\text{LCD}_D, C_D, \text{BoundTrans}_D) \wedge \text{LocalPersist}_D \wedge \text{CarrierReq}_D \Rightarrow \text{FieldRole}_D$$

where

$$\text{FieldRole}_D \equiv \exists \Phi_D : \text{Dom}(\Phi_D) \subseteq \text{LCD}_D \rightarrow \text{CarProf}_D.$$

The map Φ_D is a field-role assignment: it assigns carrier profiles to exactly those local continuation domains that participate in standing-bearing localized persistence under bounded perturbation transport.

The paper derives the minimal carrier role of physical fields. It does not derive a field equation, smooth manifold support, a gauge principle, an action functional, canonical or path-integral quantization, local algebra nets, stress-energy coupling, particle excitations, or Lorentz-covariant dynamics. These are not denied. They are later strengthening conditions. The result is role-first: any later physical field formalism that carries localized persistence under bounded perturbation transport must occupy the carrier role derived here.

2 Cumulative placement in the realized-physics arc

The first six papers in the arc separate structural theorem from physical role:

Paper	Structural theorem	Physical role
Minimal non-vacuity	realized persistence is not reconstruction-null	physical realization cannot be null
Standing localization	standing-bearing persistence is boundedly localized	origin of locality
Continuation loci	bounded loci induce continuation-neighborhood domains	origin of local-domain structure
Proto-metric representation	overlap-stable comparison induces a quotient carrier	origin of metric role
Bounded perturbation transport	perturbation burden is witness-mediated and bounded	origin of finite-propagation constraint
Fields as persistence carriers	localized persistence requires distributed carrier assignment	origin of field structure

The present paper is therefore not a detour away from physics. It identifies the structural role later occupied by physical fields. If bounded perturbation transport is the precondition for propagation, then carrier assignment is the precondition for field-mediated propagation. Without such a carrier, one has local domains and transport paths but not the structure that preserves the standing-bearing content transported along them.

Dependency 2.1 (Imported structure). The argument imports only the following layers.

- (i) NonVacRec_D : carriers cannot be null, formal-only, or reconstruction-inert.
- (ii) $[B_D]_{\equiv_B}$: carriers must be bounded-role structures rather than unbounded global media.
- (iii) $(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D)$: carriers are assigned over continuation-neighborhood domains.
- (iv) $C_D = \text{CompWit}_D / \sim_D$: carrier states preserve quotient comparison classes.
- (v) BoundTrans_D : carriers support bounded witness-mediated perturbation transport.

No primitive field substrate, spacetime base, metric tensor, or field equation is imported.

3 Definitions and type discipline

The definitions below fix the type of the result. They are intentionally weaker than standard field-theoretic structures.

Definition 3.1 (Local continuation domain). A local continuation domain $U \in \text{LCD}_D$ is an admitted domain generated by bounded standing-localization classes under boundary-compatible continuation-neighborhood paths. It is not assumed to be an open set, manifold region, causal diamond, algebraic localization region, or metric ball.

Definition 3.2 (Quotient comparison carrier). The comparison carrier C_D is the quotient of admissible comparison witnesses by equivalence of standing-bearing comparison role:

$$C_D := \text{CompWit}_D / \sim_D .$$

It is internal to the fixed regime. It need not be scalar-valued, total, linear, continuous, symmetric, or globally comparable.

Definition 3.3 (Bounded perturbation transport). $\text{BoundTrans}_D(p, U, V)$ holds when a standing-relevant perturbation burden p is carried from local continuation domain U to local continuation domain V through admitted continuation-neighborhood paths equipped with boundary-preserving transport witnesses and quotient-stable comparison-class composition.

Definition 3.4 (Persistence carrier). A persistence carrier is a domain-internal role structure whose admissible modification changes the standing-bearing transport organization of the regime. It retains standing-bearing consequence across local continuation, perturbation burden, restoration, and admitted transport. A carrier is not, by definition, a substance, particle, hidden variable, information register, spacetime point, or external medium. It is the role required for persistence to be carried rather than merely recorded.

Definition 3.5 (Carrier profile). For a local continuation domain U , a carrier profile is the tuple

$$\text{CarProf}_D(U) = (B_U, S_U, I_U, R_U, P_U, Q_U, \partial_U, T_U),$$

where the transport interface T_U mediates admissible perturbation transport and the remaining components stabilize or condition that mediation:

Component	Role
B_U	bounded support or local role locus
S_U	standing status
I_U	identity-memory profile
R_U	restoration profile
P_U	perturbation-burden profile
Q_U	quotient/comparison class in C_D
∂_U	boundary-trace interface
T_U	admitted transport interface

The tuple is not a coordinate chart, information record, or field value in the classical sense. The transport interface T_U mediates admitted transport; the remaining components stabilize that mediation by fixing local support, standing status, identity-memory, restoration, perturbation burden, quotient comparison, and boundary trace. The profile therefore carries the transport role rather than merely recording it.

Each component is inherited from a prior forced role rather than selected as a convenient data list:

Component	Carrier role	Forced by
B	bounded local support	standing localization
S	standing status	realized persistence
I	identity-memory	persistence under continuation
R	restoration profile	standing-retention under perturbation
P	perturbation-burden profile	bounded perturbation transport
Q	quotient comparison class	comparison-carrier induction
∂	boundary-trace interface	continuation-neighborhood compatibility
T	transport interface	witness-mediated bounded transport

This inheritance is important: the carrier profile is not invented to resemble a field. It collects the roles already forced by localized persistence, continuation-neighborhood coherence, quotient comparison, and bounded transport.

Definition 3.6 (Field-role assignment). A field-role assignment is a partial map

$$\Phi_D : \text{Dom}(\Phi_D) \subseteq \text{LCD}_D \rightarrow \text{CarProf}_D.$$

It is partial because not every formal local domain need participate in standing-bearing persistence or admitted perturbation transport. Totalizing Φ_D by default would import a hidden global field substrate. The assignment is field-role structure only when it mediates admissible persistence transport; a passive labelling of domains is bookkeeping and does not satisfy the definition.

Definition 3.7 (Carrier coherence). Carrier profiles are coherent along admitted restriction, overlap, and transport relations when the standing-bearing components

$$(S, I, R, P, Q, \partial, T)$$

are preserved up to the equivalences fixed by boundary trace, quotient comparison, and transport-witness structure.

4 Carrier role versus bookkeeping

A distributed assignment over local domains is not automatically a field role. It becomes field-structural only when it participates in the standing-bearing mediation of localized persistence. This distinction is essential: bookkeeping can label a transport process after the fact, but it cannot carry perturbation burden through the process.

Definition 4.1 (Carrier mediation). A carrier profile $\Phi_D(U)$ mediates admissible perturbation transport when admissible modification of the profile changes the standing-bearing transport organization involving U . Equivalently, changes in the profile can alter which perturbation burdens are preserved, which transport witnesses remain admissible, which boundary interfaces agree, or which quotient-comparison classes are retained. If all changes in an assignment leave standing status, identity-memory, restoration status, perturbation burden, quotient comparison, boundary trace, and transport interface unchanged, then the assignment is bookkeeping or skin rather than field-role structure.

Proposition 4.2 (Carrier mediation). *If a perturbation burden is admissibly transported from U to V , then the carrier profiles at U and V must participate in mediating that transport. They cannot merely describe the transport after it has occurred.*

Proof. Admissible transport preserves standing-bearing structure: source burden, target burden, boundary trace, quotient identity, restoration status, comparison class, and local-domain role. If the profiles at U and V do not participate in preserving these quantities, then the transport is carried by some other structure. That alternative is either a hidden carrier, a path label, an external channel, or an identity reset. Each case is inadmissible in the fixed regime. Hence the carrier profiles must mediate the transport role itself. $\square$

5 Transport requires carrier support

Bounded perturbation transport is stronger than route availability. A path may identify where transport is admitted; it does not itself preserve what is transported. The transported burden must remain a standing-relevant consequence rather than a label carried by notation.

Proposition 5.1 (Transport requires carrier support). *If $\text{BoundTrans}_D(p, U, V)$, then there exist carrier profiles assigned at the source and target domains whose roles preserve the transported standing-bearing burden:*

$$\text{BoundTrans}_D(p, U, V) \Rightarrow \exists \Phi_D(U), \Phi_D(V).$$

Proof. By definition of bounded perturbation transport, the burden p is not merely syntactically related to U and V . It is standing-relevant: it alters or conditions the evaluability of standing status, restoration status, boundary trace, quotient identity, comparison class, or local-domain role. Transport without carrier support would leave these roles unassigned at the source or target, or would assign them only as passive descriptions. If the roles are unassigned or passive only, the alleged transport is one of the excluded cases from the bounded-transport theorem: path syntax without standing-bearing content, hidden transfer, identity reset, external-channel import, or domain drift. Therefore admissible bounded transport requires carrier profiles at the domains whose persistence burden is being transported, and those profiles must mediate the transport role. $\square$

Remark 5.2. This proposition is the point at which Paper 6 becomes field-facing. Paper 5 supplies bounded transport paths and witnesses. Paper 6 shows that those paths and witnesses require role-bearing support that mediates, rather than merely labels, the transported standing. The support is not a primitive material medium; it is the carrier profile required for localized persistence to remain standing-bearing under transport.

6 Failure of non-field carrier candidates

The carrier role cannot be discharged by structures that are too small, too syntactic, too global, or too hidden. These failure modes explain why field structure enters as distributed carrier assignment.

Lemma 6.1 (Point-carrier insufficiency). *An isolated point or atom cannot carry localized persistence under bounded perturbation transport unless it supplies the carrier roles*

$$(B, S, I, R, P, Q, \partial, T).$$

If it supplies those roles, it is no longer a primitive point; it is a carrier profile.

Proof. A primitive point can be named or selected, but naming is not standing-bearing carriage. Bounded transport requires identity-memory, restoration, boundary trace, quotient comparison, perturbation burden, and transport interface. A point lacking those roles cannot preserve the transported burden. A point equipped with those roles functions as the tuple $\text{CarProf}_D(U)$, which is exactly a local carrier profile. Thus point ontology either fails to carry localized persistence or becomes field-role structure in local form. $\square$

Lemma 6.2 (Path-label collapse). *A transport path without carrier profiles records route syntax only. It does not carry standing-bearing persistence.*

Proof. A path $U_0 \text{Adj}_D \cdots \text{Adj}_D U_n$ gives admitted neighborhood order. The path does not by itself store standing status, restoration profile, identity-memory, perturbation-burden profile, quotient class, or boundary-trace interface. If these roles are absent, the path label can describe possible transport but cannot bear transported standing. If these roles are added at the domains along the path, the path is supplemented by carrier profiles and therefore no longer functions as a carrier by itself. $\square$

Lemma 6.3 (Global carrier collapse). *A global undifferentiated substrate cannot serve as the carrier of localized persistence unless it factors through local continuation domains. If it does not factor, it collapses standing-verdict separability and bounded perturbation transport. If it does factor, it is a distributed carrier assignment rather than an undifferentiated global substrate.*

Proof. Suppose a single global substrate carries all persistence without local factorization. Then perturbation burden cannot be assigned to bounded local domains without becoming immediately global, and the separability results inherited from continuation-neighborhood structure collapse. If local perturbation burden, boundary traces, and restoration status remain distinguishable, the substrate has already factored into domain-indexed carrier profiles. Hence the global substrate either fails to preserve localized persistence or realizes the field-role assignment it was meant to avoid. $\square$

7 Minimal field role requirement

The preceding failures identify the minimal field role. A field role is not imposed because field theory is familiar; it is forced because any successful carrier must mediate persistence transport through the following structure.

Proposition 7.1 (Minimal field role requirement). *Any carrier of localized persistence under bounded perturbation transport must preserve bounded local support, standing status, identity-memory, restoration profile, perturbation-burden profile, quotient/comparison class, boundary-trace interface, and transport interface. Thus*

$$\text{CarrierReq}_D \Rightarrow (B, S, I, R, P, Q, \partial, T).$$

Proof. Omit bounded local support and the carrier becomes global or unbounded, contradicting standing localization. Omit standing status and it no longer carries persistence rather than description. Omit identity-memory and transport becomes identity reset. Omit restoration profile and perturbation burden cannot be evaluated against continuation. Omit perturbation-burden profile and there is nothing standing-relevant to transport. Omit quotient comparison and transport loses the comparison class fixed by Paper 4. Omit boundary trace and carrier interfaces become selector imports. Omit transport interface and bounded perturbation transport cannot be mediated. The interface T is therefore not an appended label; it is the carrier-facing expression of bounded perturbation transport, while the other components determine what the interface is allowed to preserve. Each component is necessary for the field role. $\square$

Remark 7.2 (Why this is a field role). The role is field-structural because it assigns carrier profiles locally over continuation domains and constrains their compatibility across restriction, boundary overlap, and transport. It is not field-structural merely because profiles are distributed; it is field-structural because those profiles mediate standing-bearing persistence. That is the structural content common to later field species before one specifies values, equations, symmetries, manifolds, or quantization.

8 Carrier induction from transport coherence

The field-role assignment is induced by the transport problem rather than stipulated as a convenient representation. Localized persistence under bounded perturbation transport already fixes where carrier profiles must exist and what roles they must preserve.

Theorem 8.1 (Carrier induction from transport coherence). *If localized persistence is preserved under bounded perturbation transport inside a fixed admissible regime, then transport-support-free persistence mediation is impossible. The required carrier profiles therefore assemble into a partial field-role assignment over the participating local continuation domains:*

$$(\text{LCD}_D, C_D, \text{BoundTrans}_D) \wedge \text{LocalPersist}_D \Rightarrow \exists \Phi_D \text{ on the transport-participating domains.}$$

The induced assignment is unique only up to admissible carrier-role equivalence: two candidate profiles are equivalent as carriers exactly when they preserve the same standing-bearing transport, restoration, boundary, and quotient-comparison behavior. No scalar value space, field equation, or global substrate is implied.

Proof. For each admitted transport instance $\text{BoundTrans}_D(p, U, V)$, the transport support and mediation propositions exclude transport-support-free persistence: a perturbation burden cannot be preserved by path syntax alone, by an external record, or by a hidden transfer. Carrier profiles are therefore required at the participating source and target domains. For multi-step transport, quotient-stable composition supplies admitted intermediate domains and transport witnesses. Carrier mediation requires the profiles on those domains to participate in the same standing-bearing transport organization, while carrier coherence requires agreement under boundary, quotient, and restoration equivalence. Profiles preserving exactly the same standing-bearing transport, restoration, boundary, and quotient-comparison behavior are indistinguishable as carriers inside the fixed regime. Collecting exactly the domains for which such profiles are required gives a partial domain $\text{Dom}(\Phi_D) \subseteq \text{LCD}_D$. Assigning to each such domain its required carrier profile defines Φ_D . The construction is partial because domains not participating in standing-bearing persistence or admitted perturbation transport are not assigned by default; totalizing the assignment beyond those domains would import a hidden global field substrate. Thus Φ_D is induced as the minimal carrier assignment required by transport coherence rather than stipulated as a field-theoretic primitive. $\square$

9 Field-carrier role theorem

Theorem 9.1 (Field-carrier role theorem). *Let D be a fixed admissible realized persistence regime with local continuation domains LCD_D , quotient comparison carrier C_D , and bounded perturbation transport BoundTrans_D . If localized persistence in D must retain standing-bearing consequence under admissible perturbation transport, then a distributed field-role assignment over local continuation domains is induced by the impossibility of transport-support-free persistence mediation:*

$$(\text{LCD}_D, C_D, \text{BoundTrans}_D) \wedge \text{LocalPersist}_D \wedge \text{CarrierReq}_D \Rightarrow \text{FieldRole}_D.$$

Equivalently,

$$\text{FieldRole}_D \equiv \exists \Phi_D : \text{Dom}(\Phi_D) \subseteq \text{LCD}_D \rightarrow \text{CarProf}_D$$

preserving $(S, I, R, P, Q, \partial, T)$ under admitted restriction, overlap, and transport.

Proof. Localized persistence requires retention of standing-bearing consequence within bounded local domains. Bounded perturbation transport requires source and target burden preservation under admitted transport witnesses. By the transport support and mediation propositions, such burden preservation requires carrier profiles at the domains that participate in transport, and those profiles must mediate rather than merely record the transport. By carrier induction from transport coherence, the profiles are fixed up to equality of admissible carrier role. By the minimal field role requirement, each carrier profile must contain bounded support, standing status, identity-memory, restoration profile, perturbation-burden profile, quotient comparison class, boundary interface, and transport interface. Because perturbation transport occurs across local continuation domains rather than at a single isolated atom, these profiles are induced throughout the participating domain structure rather than externally stipulated. Because transport is boundary- and quotient-sensitive, the profiles must cohere under admitted restriction, overlap, and transport. By carrier mediation, this induced assignment is not bookkeeping: its admissible modification changes the transport organization. This coherent assignment is exactly a field-role assignment Φ_D , and hence FieldRole_D follows. $\square$

Remark 9.2. The theorem derives fields at the level of role necessity. It does not choose a value space, differentiable base, fiber, section, operator algebra, gauge group, action, or equation of motion. Those are later specializations of the role just derived.

10 Restriction and local compatibility

The field role is not merely a list of local profiles. It must restrict coherently to admitted subdomains, otherwise local persistence would depend on arbitrary domain presentation.

Definition 10.1 (Carrier restriction). For an admitted local continuation subdomain relation $V \preceq_{\text{LCD}} U$, a carrier restriction is a partial operation

$$\text{res}_{U,V} : \Phi_D(U) \rightarrow \Phi_D(V)$$

defined only when V is a legitimate continuation subdomain of U and the standing-bearing components remain boundary- and quotient-compatible.

Proposition 10.2 (Restriction compatibility). *If $V \preceq_{\text{LCD}} U$, then the carrier profile assigned to V must agree with the restriction of the profile assigned to U on all standing-bearing role components:*

$$\text{res}_{U,V}(S_U, I_U, R_U, P_U, Q_U, \partial_U, T_U) = (S_V, I_V, R_V, P_V, Q_V, \partial_V, T_V)$$

up to the admissible equivalences already fixed by boundary traces and quotient comparison.

Proof. If the restricted carrier profile disagrees on standing status, identity-memory, restoration, perturbation burden, quotient class, boundary trace, or transport interface, then persistence in the subdomain becomes a presentation artifact rather than the restriction of the larger domain's carrier role. Such disagreement is either boundary mismatch, quotient drift, selector import, or identity reset. Therefore restriction compatibility is required. $\square$

Remark 10.3. This is restriction-like behavior, not a primitive sheaf assumption. No sheaf-theoretic completeness, covering, gluing, cohomological, bundle, or section axiom is assumed. The restriction maps enforce only local carrier compatibility under admissible continuation. A sheaf or bundle formulation may become appropriate after additional mathematical structure is supplied, but the present result does not assume one.

11 Transport compatibility

Carrier profiles must also be compatible with admitted transport witnesses. This is where the transport interface T_U becomes central: it mediates the admitted passage of standing-bearing burden, while $(S, I, R, P, Q, \partial)$ stabilize what is being mediated.

Proposition 11.1 (Carrier transport compatibility). *If $\tau_D(U, V, p)$ is an admitted perturbation-transport witness, then carrier profiles must satisfy an admissible transport relation*

$$\Phi_D(U) \xrightarrow{\tau_D} \Phi_D(V)$$

preserving $(S, I, R, P, Q, \partial, T)$. If no such relation exists, the perturbation transport is not carried inside D .

Proof. A transport witness is admitted only when source burden, target burden, boundary trace, quotient identity, restoration status, comparison class, and local-domain role are preserved. These are exactly the field-profile components that make localized persistence transportable. If the source and target carrier profiles do not stand in a preservation relation induced by τ_D , then the witness cannot carry the burden between them. The alleged transport is then either route syntax, hidden transfer, or domain drift. Hence carrier transport compatibility is necessary. $\square$

12 Boundary compatibility and overlap coherence

The carrier assignment must also cohere where local domains overlap or meet through boundary-compatible adjacency.

Proposition 12.1 (Carrier boundary coherence). *If $U, V \in \text{Dom}(\Phi_D)$ overlap or are adjacent through admitted boundary-compatible continuation, then $\Phi_D(U)$ and $\Phi_D(V)$ must agree on the boundary-trace interface relevant to transported standing. Otherwise transport across the boundary is inadmissible.*

Proof. Boundary-compatible continuation requires jointly reconstructible boundary traces. If carrier profiles disagree on the boundary interface that fixes standing-bearing transport, a perturbation transferred across the boundary would either alter identity-memory, shift quotient class, reset restoration status, or import a selector to choose which boundary profile governs. Each option violates fixed-domain admissibility. Therefore boundary coherence is required for field-role assignment. $\square$

Remark 12.2 (Physical reading). Boundary coherence is the field-facing analogue of local compatibility. It is not yet gauge structure. Gauge structure enters when multiple carrier presentations preserve the same boundary-fixed field role and admissible transformations among them become necessary.

13 No primitive field substrate

Primitive field substrates are unnecessary once carrier mediation has been induced from localized persistence and bounded transport. The issue is not whether familiar field formalisms are useful; it is whether a medium may be imported before the carrier role has been fixed. The answer is no: either the proposed substrate performs the derived carrier role, or it does not carry localized persistence.

Theorem 13.1 (Primitive field substrate redundancy). *A field substrate need not be admitted as primitive prior to carrier-role fixation. If a proposed substrate preserves $(S, I, R, P, Q, \partial, T)$ over local continuation domains, then it is a realization of the derived field role. If it does not preserve those components, it cannot carry localized persistence under bounded perturbation transport.*

Proof. A proposed primitive substrate with no specified carrier-mediation role is either global, hidden, trace-only, or formal. If it is global and undifferentiated, the global carrier collapse lemma applies. If it is hidden, it cannot ground realized persistence unless it is made accessible through admissible boundary traces and transport witnesses. If it is trace-only or formal, it lacks standing-bearing content. If, on the other hand, it supplies bounded support, standing status, identity-memory, restoration, perturbation burden, quotient comparison, boundary trace, and transport interface, then it instantiates Φ_D . Thus the primitive-substrate alternative either fails to carry persistence or becomes the derived field role. $\square$

14 Skin-only and hidden carrier failures

Lemma 14.1 (Skin-only field collapse). *A field assignment that changes only notation, diagram placement, coordinate labels, or presentational skin carries no standing-bearing persistence. A physical field role must operate at the role level, not merely at the skin level.*

Proof. The Anchor-Tensor-Skin decomposition separates presentational surface from role-bearing structure [6]. If a field assignment changes only presentation, then standing status, perturbation burden, restoration profile, quotient comparison, and transport interface remain unaffected. Such an assignment cannot carry localized persistence. It is skin, not field role. $\square$

Corollary 14.2 (Hidden carrier reservoir exclusion). *A latent carrier structure inaccessible to admissible boundary traces, transport witnesses, and quotient comparison cannot carry realized persistence. If it becomes accessible, it belongs to Φ_D . If it remains inaccessible, it is realization-inert.*

Proof. Realized persistence requires domain-internal standing-bearing accessibility. A hidden reservoir that never enters boundary traces, transport witnesses, or quotient comparison cannot preserve or alter carrier profiles. If it does enter those structures, it is no longer hidden and is part of the field-role assignment. $\square$

15 Relation to standard physical fields

The result identifies what later field theories must do, not what they lack. Physical fields in standard frameworks instantiate distributed carrier support: they assign local physical content, mediate interactions or transport, preserve continuity of that content across regions, impose boundary compatibility, and provide the structures whose excitations or modes may later be particle-like. Classical field theory adds manifold support, value spaces, and equations; gauge theory adds presentation equivalences and connections; quantum field theory adds operator or algebraic structure; particle descriptions introduce excitations or localized persistence classes [7, 14, 13, 12, 10, 8, 9].

Standard field ingredient	Status in the present paper
Base manifold or spacetime support	not assumed; local continuation domains are prior
Scalar, vector, spinor, or tensor values	later value-space specialization
Differential field equation	later dynamics
Lagrangian or Hamiltonian	later variational or dynamical structure
Gauge redundancy or gauge connection	later boundary-presentation equivalence structure
Quantum operator field	later quantization or algebraic specialization
Stress-energy or gravitational coupling	later mass/inertia/gravity layer
Particle excitations	later localized persistence-class analysis

This hierarchy prevents both overclaiming and underclaiming. The paper does not derive Maxwell, Yang-Mills, Dirac, Klein-Gordon, or quantum operator fields. But it does derive the

carrier role those fields occupy when they function as physical fields: they locally support, mediate, and stabilize standing-bearing persistence and perturbation transport. Later field theories can add value spaces, equations, symmetries, and quantization only after this carrier role is available.

16 Gauge structure, particles, and hidden-variable cautions

Gauge structure is downstream. A carrier assignment becomes gauge-facing only when multiple presentations of $\Phi_D(U)$ preserve the same standing-bearing role and boundary traces fix admissible transformations among those presentations. That is not needed for the field-carrier role itself. The present paper supplies what later gauge structure must act on: boundary-compatible carrier profiles preserving standing, identity, restoration, burden, quotient comparison, and transport interface.

Particles are also downstream. The present paper does not treat particles as primitive carriers. A later particle paper can treat particle identity as stable localized persistence classes or excitation-like modes of carrier structure. That analysis requires field-role structure first; otherwise a particle would have to be either point-only, path-only, hidden, or externally imported.

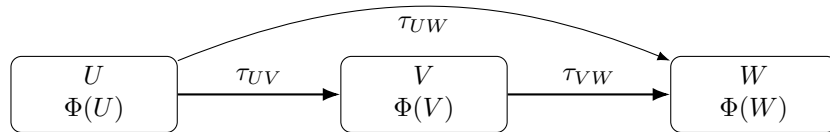
The field role derived here is not a local hidden-variable field. It does not assert deterministic local states, Bell-local factorization, common-cause screening-off, outcome determination, or a subquantum signal medium [15, 16]. It asserts only that realized localized persistence under bounded perturbation transport requires carrier profiles internal to the admissible continuation regime.

17 Finite witness model

A finite witness model helps distinguish field-role structure from point, path, or global-substrate substitutes. Let

$$G_D = (V, E, \Phi, \tau, C_D)$$

where V are local continuation domains, E are admitted adjacency or transport relations, $\Phi(v)$ is a carrier profile at v , τ_e is a perturbation-transport witness on edge e , and C_D is the quotient comparison carrier.



same-role transport requires, where defined,

$$[\tau_{UW}] = [\tau_{UV}] \odot_D [\tau_{VW}]$$

The model exhibits the following alternatives:

- (i) no Φ : transport has no carrier;
- (ii) point-only Φ : boundary, restoration, quotient, or transport roles are missing;
- (iii) global constant Φ : local separability collapses unless it factors;

- (iv) arbitrary Φ : carrier selection is imported;
- (v) path labels without Φ : transport is skin-only route syntax;
- (vi) incompatible overlaps: quotient stability fails;
- (vii) coherent Φ with boundary and transport compatibility: field-role mediation is realized.

The finite graph is a witness model only. It instantiates one admissible realization of carrier mediation; it does not define the general carrier structure, define fields as graph functions, or assume graph-theoretic ontology.

18 Carrier route exhaustion

Theorem 18.1 (Carrier route exhaustion). *Within a fixed admissible realized persistence regime satisfying the dependencies of Papers 1-5, any proposed carrier of localized persistence under bounded perturbation transport falls into one of the following classes:*

- (i) mediated distributed field-role assignment;
- (ii) bookkeeping-only assignment;
- (iii) point-only or path-only assignment;
- (iv) global substrate assignment;
- (v) hidden or default assignment;
- (vi) transport-free, trace-only, or skin-only assignment;
- (vii) identity-reset or cross-domain assignment.

Only the first class is admissible as a carrier of localized persistence under bounded perturbation transport.

Proof. Bookkeeping-only assignments fail by the carrier mediation proposition: they describe but do not carry standing-bearing transport. Point-only carriers fail by point-carrier insufficiency, and path-label carriers fail by path-label collapse. Global undifferentiated substrates fail by global carrier collapse unless they factor into local carrier profiles, in which case they instantiate the first class. Hidden or default carriers either become accessible through boundary traces and transport witnesses, hence enter Φ_D , or remain realization-inert. Trace-only, skin-only, or transport-free carriers carry no standing-bearing transport role. Identity-reset carriers destroy persistence, and cross-domain imports exit the fixed domain. The only remaining admissible class is the mediated distributed field-role assignment. $\square$

19 Normal form

The result may be stated in one line:

$$\boxed{(\text{LCD}_D, C_D, \text{BoundTrans}_D) \wedge \text{LocalPersist}_D \wedge \text{CarrierReq}_D \Rightarrow \text{FieldRole}_D}$$

with

$$\text{FieldRole}_D \equiv \exists \Phi_D : \text{Dom}(\Phi_D) \subseteq \text{LCD}_D \rightarrow \text{CarProf}_D$$

and

$$\text{CarProf}_D(U) = (B_U, S_U, I_U, R_U, P_U, Q_U, \partial_U, T_U).$$

Thus field structure enters at the exact point where localized persistence under bounded perturbation transport requires domain-distributed carrier mediation. The field role is neither imported as a primitive medium nor postponed until a completed field theory is already in hand.

20 Conclusion

Paper 1 established that realized persistence cannot be reconstruction-null. Paper 2 showed that standing-bearing persistence localizes. Paper 3 derived continuation-neighborhood domains. Paper 4 derived quotient-stable comparison carriers. Paper 5 established bounded perturbation transport. The present paper shows that such transport requires a distributed carrier assignment over local continuation domains. Isolated points, path labels, global undifferentiated substrates, hidden reservoirs, identity-reset carriers, cross-domain imports, and trace-only descriptions cannot carry localized persistence under bounded transport.

What has been isolated is the minimal distributed mediation structure required for localized persistence under bounded perturbation transport: a boundary-compatible, transport-compatible carrier assignment preserving standing status, identity-memory, restoration status, perturbation burden, quotient comparison, boundary trace, and local-domain transport interface. Classical, quantum, gauge, tensor, and manifold-based fields are not assumed here; they are later specializations of the field role derived here. In this sense AASC enters real physics not by postulating field substance, but by deriving the persistence-mediation role that physical fields occupy.

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